

Stochastic Calculus & Finance: The Heath-Jarrow-Morton Model

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where $f(t, T)$ is the instantaneous rate for time $T > t$ that can be locked in at time t . This is the Heath-Jarrow-Morton model, meant to model forward rates.

Below is the rough ordering of topics that will be covered:

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- Measures
- Sigma-Algebras, Random Variables, and Information
- Random Walks and Brownian Motion
- Putting it all Together

The Canonical Example

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For example, one element of this space would be

$$(1, 1, 1, \dots),$$

which represents a countably infinite chain of heads.

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In particular, a probability measure $\mathbb{P}$, requires that $\mathbb{P}(\Omega) = 1$.

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Definition 2

Formally, let Ω be a nonempty set, and $\mathcal{F}$ a collection of subsets of Ω . $\mathcal{F}$ is a sigma-algebra if:

1. $\emptyset \in \mathcal{F}$
2. $\forall A \subseteq \Omega$, s.t. $A \in \mathcal{F} \implies A^c \in \mathcal{F}$
3. $A_1, A_2, \dots \in \mathcal{F} \implies \bigcup_{n=0}^{\infty} A_n \in \mathcal{F}$

Intuitively, $\mathcal{F}$ is designed to contain all events whose probabilities make sense in our model.

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Together, the triple

$$(\Omega, \mathcal{F}, \mathbb{P})$$

is called a probability space, and it is the backbone of probability theory.

A random variable is a function from the outcome space to the real numbers:

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So if

$$\omega = (1, -1, 1, 1, \dots),$$

then

$$X_1(\omega) = 1, \quad X_2(\omega) = -1.$$

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The sequence

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is called a filtration. A filtration is the mathematical model of “what we know so far.”

Definition 3

Formally, a filtration is a time-indexed sequence of σ -algebras $\mathcal{F}(t)$, $t \in [0, T]$ on a nonempty Ω , such that, for $s \leq t$,

$$\mathcal{F}(s) \subseteq \mathcal{F}(t)$$

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A model should not depend on future information. So quantities like

$$f(t, T), \quad \sigma(t, T), \quad W(t)$$

must be based only on information available at time t .

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So S_n records the cumulative position after n random steps.

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Thus, we must adapt this process to make use of it for our modeling purposes.

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Definition 5

Fix some $n \in \mathbb{Z}^+$. Then, the scaled symmetric random walk is defined as

$$W^{(n)}(t) = \frac{1}{\sqrt{n}} S_{nt},$$

where S_{nt} is a regular symmetric random walk, provided $nt \in \mathbb{Z}$.

The Transition to Continuous (cont.)

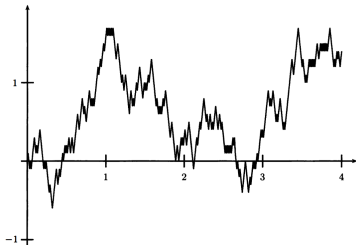


Fig. 3.2.2. A sample path of $W^{(100)}$.

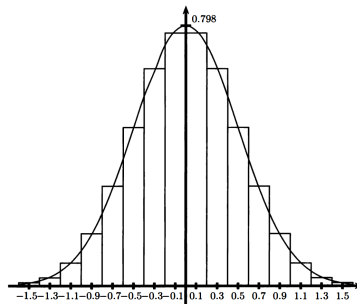


Fig. 3.2.3. Distribution of $W^{(100)}(0.25)$ and normal curve $y = \frac{2}{\sqrt{2\pi}} e^{-2x^2}$.

The Transition to Continuous (cont.)

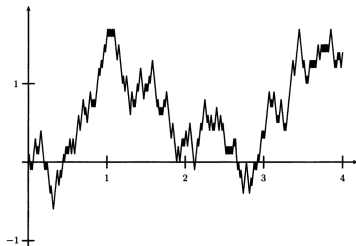


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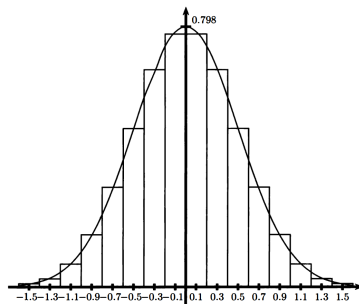


Fig. 3.2.3. Distribution of $W^{(100)}(0.25)$ and normal curve $y = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$.

As we make the increments and time steps smaller, the path becomes increasingly jagged, with increments approaching a normal distribution.

We then take the limiting object of these scaled random walks (proven rigorously by taking the limit as $n \rightarrow \infty$ and using CLT.)

Obtaining the Brownian Motion

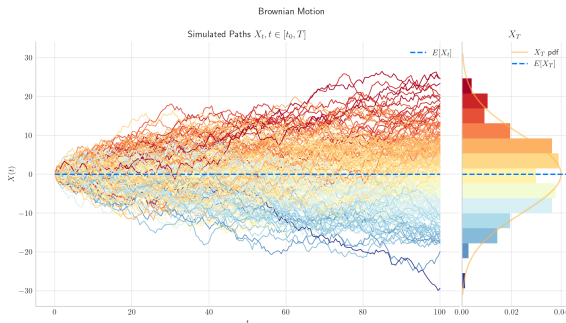
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Source: quantgirluk.github.io/Understanding-Quantitative-Finance

Definition 6

Formally, a continuous function $W(t)$ dependent on $\omega \in \Omega$ is a Brownian Motion if, for all $0 = t_0 < \dots < t_m$, the increments

$$W(t_1) - W(t_0), \dots, W(t_m) - W(t_{m-1})$$

are independent and normally distributed. In addition,

$$E[W(t_{i+1}) - W(t_i)] = 0,$$

$$\text{Var}[W(t_{i+1}) - W(t_i)] = t_{i+1} - t_i.$$

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Although expressing the solution to f is possible in terms of an ordinary Lebesgue integral and an Ito Integral (for the Brownian Motion), in practice, numerical methods are often needed for evaluation.

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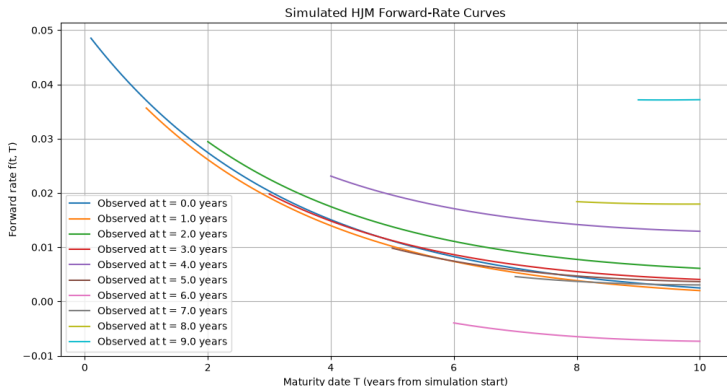
Specifically, we need observations of the forward rate for various different times in the past for various different relative maturities. In addition, we also require incremental observations for some time offset δ .

Technically, this only gives us enough information to estimate σ , but through a technique called change of measure, we do not require α .

A Coded Example

Below is a coded implementation of the HJM model on a dummy initial forward rate curve

$$f(0, T) = 0.05e^{-0.3T}.$$



- [1] Steven E. Shreve. *Stochastic Calculus for Finance II: Continuous-Time Models*. Springer Finance, Springer, 2004.